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Functional equations

Level 4 Olympiad · Algebra

How to find every function satisfying an equation: choosing substitutions, injectivity and surjectivity, Cauchy's equation over the rationals, and complete write-ups that check the answer.

By the end you can: You will be able to choose substitutions that pin down special values and make awkward terms vanish; You will be able to prove and use injectivity and surjectivity, and solve Cauchy's equation over the rationals; You will be able to write a complete 'find all functions' solution, avoiding the pointwise trap and checking every answer.

Lesson

Consider the following problem. Find every function $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) + 2f(1 - x) = x^2 \quad \text{for all real } x.$$

The equation ties the value at x to the value at $1 - x$, so on its own it does not tell us $f(x)$. But it holds for *every* real number, so it also holds with x replaced by $1 - x$:

$$f(1 - x) + 2f(x) = (1 - x)^2.$$

Now we have two linear equations in the two unknowns $f(x)$ and $f(1 - x)$. Doubling the second and subtracting the first gives $3f(x) = 2(1 - x)^2 - x^2$, so

$$f(x) = \frac{x^2 - 4x + 2}{3}.$$

Note that so far we have shown only that *if* a solution exists, it must be this function. Substituting back, $f(1 - x) = \frac{1}{3}(x^2 + 2x - 1)$ and $f(x) + 2f(1 - x) = \frac{1}{3}(3x^2) = x^2$, so it is a solution, and it is the only one. Every functional equation write-up has this two-part structure: "any solution must be this", followed by "and this works".

What "find all functions" asks for

A complete answer has two halves. **Necessity:** assuming f satisfies the equation, deduce that f is one of an explicit list. **Sufficiency:** check that every function on the list satisfies the equation for *all* allowed values of the variables. At BMO a missing check costs marks, and occasionally a candidate you derived fails the check, which is why it is required.

The domain is part of the problem. The same formula can have different answers on $\mathbb{R}$, on $\mathbb{Q}$ and on the positive integers, and a substitution like $x = 0$ is illegal if 0 is not in the domain. Read the first line of the problem twice.

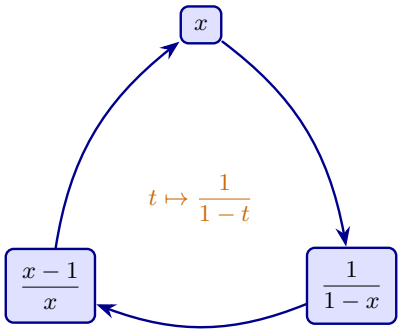
The first substitutions

An equation that holds for all x and y gives one fact for each choice of x and y, and the work lies in choosing values that give useful facts. The standard first substitutions are as follows.

1. Set variables to 0 (or to 1 in a multiplicative equation) to find $f(0)$ or $f(1)$.
2. Try $y = x$ and $y = -x$; if the equation is not symmetric, swap x and y and compare.

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3. Choose one variable in terms of the other so that an awkward argument becomes a value you know, or so that two arguments become equal.
4. Replace x by an expression whose repeated application cycles back to x, such as $1-x$ (a cycle of length two, as in the opening problem) or $\frac{1}{1-x}$ (a cycle of length three).



With a cycle of length three you substitute twice and get three equations in three unknowns, which you then solve exactly as above.

Example 1. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x-y) = f(x) + f(y) - 2xy$ for all real x, y.

Solution.
The term $-2xy$ looks like part of $(x-y)^2$, so we guess $f(x) = x^2$. Next we look for a substitution that makes the argument of $f(x-y)$ constant. Putting $y = x$ makes the argument 0, so first we need $f(0)$.
With $x = y = 0$ the equation reads $f(0) = 2f(0)$, so $f(0) = 0$. With $y = x$ it reads $f(0) = 2f(x) - 2x^2$, so $f(x) = x^2$ for every real x. That is necessity. For sufficiency, $(x-y)^2 = x^2 + y^2 - 2xy$ for all x, y, so $f(x) = x^2$ is the unique solution.
In general, the first substitution to look for is one that makes an argument constant (here $y = x$ makes $x - y = 0$).

Injective, surjective, and making terms vanish

A function is **injective** if $f(a) = f(b)$ forces $a = b$, and **surjective** if every value in the codomain is taken. Both are proved from the equation, never assumed. Both most often come from a relation like $f(f(x)) = x + c$: if $f(a) = f(b)$ then applying f gives $a + c = b + c$, so f is injective, and since $x + c$ takes every real value, f is surjective.

Injectivity lets you strip an f off both sides: if you can arrange $f(A) = f(B)$, then $A = B$. Surjectivity lets you replace "for all y, a statement about $f(y)$" by "for all real t, the same statement about t".

Example 2. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f\big(xf(y) + y\big) = xy + f(y)$ for all real x, y.

Solution.
Guessing $f(x) = ax + b$ and comparing coefficients gives $a^2 = 1$ and $b = 0$, so we expect $f(x) = x$ and $f(x) = -x$. It remains to prove that nothing else works.
Put $y = 0$: $f\big(xf(0)\big) = f(0)$ for all x. If $f(0) = c \neq 0$, then xc runs through every real number, so f is constant, equal to c; but then the equation says $c = xy + c$ for all x, y, which is false. Hence $f(0) = 0$.
Next we make the left-hand argument vanish: if $f(y) \neq 0$, take $x = -y/f(y)$ to get $0 =$

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$-y^2/f(y) + f(y)$, so $f(y)^2 = y^2$. Note that this does not yet prove $f(y) = y$ for all y or $f(y) = -y$ for all y: we only know that *for each* y the value is y or $-y$, and the sign could change from one y to another (the pointwise trap). Ruling out such mixtures is the hard part of the problem.
First, f has no zeros except 0: if $f(b) = 0$, then putting $y = b$ gives $f(b) = xb + f(b)$ for all x, so $b = 0$. In particular $k = f(1) \neq 0$. Now fix $y = 1$:

$$f(kx + 1) = x + k \quad \text{for all real } x.$$

Since $k \neq 0$, as x runs over $\mathbb{R}$ so does $t = kx + 1$, and $x = (t - 1)/k$. Hence $f(t) = \frac{t-1}{k} + k$ for every real t: the function is linear. Now $f(0) = 0$ gives $k^2 = 1$, so $f(t) = t$ or $f(t) = -t$.
Both work: for $f(t) = t$ each side is $xy+y$, and for $f(t) = -t$ the left side is $f(-xy+y) = xy-y$, which equals the right side $xy-y$. In general, fixing one variable can make the equation describe f along a whole line; if that line covers the domain, f is determined.

Cauchy's equation over the rationals

Theorem (Cauchy over $\mathbb{Q}$). If $f : \mathbb{Q} \to \mathbb{R}$ satisfies $f(x + y) = f(x) + f(y)$ for all rationals x, y, then $f(x) = cx$ for all rational x, where $c = f(1)$.
Proof. Putting $x = y = 0$ gives $f(0) = 0$, and $y = -x$ gives $f(-x) = -f(x)$. Since $f\big((n + 1)x\big) = f(nx)+f(x)$, induction gives $f(nx) = nf(x)$ for every positive integer n. So $f(m) = mc$ for integers m, and for $n > 0$ we have $nf\big(\frac{m}{n}\big) = f(m) = mc$, so $f\big(\frac{m}{n}\big) = c \cdot \frac{m}{n}$.
On $\mathbb{R}$. Additivity alone is not enough: there exist additive functions on $\mathbb{R}$ (built using the axiom of choice) that are not of the form cx, so an extra condition is needed. The most convenient is positivity: if f is additive on $\mathbb{R}$ and $f(x) \geq 0$ whenever $x \geq 0$, then $f(x) = cx$. Indeed, for $a < b$ we get $f(b) = f(a) + f(b - a) \geq f(a)$, so f is increasing; for real x and rationals $p < x < q$ we get $cp \leq f(x) \leq cq$, and letting $p, q \to x$ gives $f(x) = cx$. Monotonicity, continuity, or boundedness on an interval would also suffice.

Example 3. Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that $f\big(x + f(y)\big) = f(x) + 4y$ for all rationals x, y.
Solution.
Trying $f(x) = kx$ gives $k^2 = 4$, so we expect $f(x) = 2x$ and $f(x) = -2x$. No substitution computes f directly, so we first prove that f is injective and surjective.
Put $x = 0$: $f\big(f(y)\big) = c + 4y$, where $c = f(0)$. The right-hand side takes each rational value exactly once as y varies, so f is injective and surjective. Put $y = 0$: $f(x + c) = f(x)$, and injectivity gives $x + c = x$, so $c = 0$ and $f\big(f(y)\big) = 4y$.
Next we prove that f is additive. The original equation contains $f(y)$ inside; replace y by $f(y)$ to get $f\big(x + f(f(y))\big) = f(x) + 4f(y)$, that is,

$$f(x + 4y) = f(x) + 4f(y).$$

Putting $x = 0$ gives $f(4y) = 4f(y)$, so $f(x + 4y) = f(x) + f(4y)$. Since $4y$ is an arbitrary rational, f is additive, and Cauchy over $\mathbb{Q}$ gives $f(x) = kx$ with $k = f(1)$ rational. Then $4y = f(f(y)) = k^2y$, so $k = \pm 2$.
Check: if $f(x) = 2x$, the left side is $2x+4y$; if $f(x) = -2x$, the left side is $f(x-2y) = -2x+4y$. Both equal the right side. If the 4 were replaced by 2, the same argument would force $k^2 = 2$, which has no rational solution, so there would be no solutions at all on $\mathbb{Q}$.

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Common traps.

- **The pointwise trap.** Showing that for each x, $f(x) = x$ or $f(x) = -x$, does not show that f is one of those two functions. You must rule out mixtures.
- **No final check,** or a check with a few special values only. The check must be for all values.
- **Dividing by something that might be zero,** such as $f(y)$ or x. Deal with the zero case separately, as in Example 2.
- **Cauchy on $\mathbb{R}$ without justification.** Additivity on $\mathbb{R}$ gives cx only with an extra condition, which you must prove.
- **Assuming the form.** "Let $f(x) = ax + b$ and compare coefficients" finds candidates; it proves nothing about other functions.
- **Ignoring the domain,** for example substituting 0 when f is defined only on positive integers.

How to spot this in a contest

At BMO Round 1 a functional equation usually appears as a "find all functions" problem near the end of the paper, typically with one or two solutions, often linear. I would approach one in the following order.

1. Guess the answers first (constants and $ax + b$), since knowing the target tells you which facts to aim for.
2. Find $f(0)$, $f(1)$ and other special values.
3. Look for an $f(f(x))$ relation, which signals injectivity or surjectivity.
4. Try to make arguments equal, or make them vanish.
5. If additivity appears, check the domain before quoting Cauchy.

On integer domains, think induction, parity and size.

In the AIME the function is usually given by a few rules and you must compute one value. The same substitutions apply, together with spotting a period or a familiar function that satisfies the rules, but remember that a guessed function is only safe if the rules force it.

Practice

Try each problem before turning to the hints and solutions at the back.

1. $\star$ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) + x\,f(1 - x) = 2x - x^2$ for all real x.

2. $\star\star$ A function f is defined for all real x other than 0 and 1, and satisfies $f(x) + f\left(\dfrac{1}{1-x}\right) = x$ for all such x. Find $f(3)$.

3. $\star\star$ Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that $f(x + y) = f(x) + f(y) + xy(x + y)$ for all rationals x, y.

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4. ⋆⋆⋆ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\big(xf(y) + f(x)\big) = f(x)f(y) + x$$

for all real x, y.

5. ⋆⋆⋆ Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that $f\big(f(n)\big) = n + 2$ for every integer n.

6. ⋆⋆⋆⋆ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 + y) = f(x)^2 + f(y)$$

for all real x, y.

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. AIME 1984 Problem 7 ⋆⋆
2. AIME 1984 Problem 12 ⋆⋆
3. AIME 1988 Problem 8 ⋆⋆
4. AIME 1997 Problem 12 ⋆⋆⋆
5. BMO Round 1 2001/02 Problem 5 ⋆⋆⋆
6. BMO Round 1 2009/10 Problem 5 ⋆⋆⋆
7. BMO Round 1 2007/08 Problem 6 ⋆⋆⋆
8. BMO Round 1 2019/20 Problem 6 ⋆⋆⋆⋆
9. BMO Round 1 2014/15 Problem 6 ⋆⋆⋆⋆
10. USAMO 2002 Problem 4 ⋆⋆⋆⋆

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Hints

1. Replace x by $1-x$ and solve the two linear equations. Before you divide, check that the coefficient can never be zero.
2. Apply the substitution $x \mapsto \frac{1}{1-x}$ to 3, then to the result, then again. Where do you end up?
3. Find one solution of the form kx^3, subtract it, and see what equation the difference satisfies.
4. Show that $f(0) = 0$; then $y = 0$ tells you something strong about $f(f(x))$. After that, fix $x = 1$.
5. Work out $f\big(f(f(n))\big)$ in two different ways. Then the values $f(0)$ and $f(1)$ determine everything, and parity matters.
6. Find $f(0)$, show that f is odd and then additive. To finish, ask what sign f takes on non-negative numbers.

Solutions

Problem 1

Answer: $f(x) = x$

Replacing x by $1-x$ gives $f(1-x)+(1-x)f(x) = 2(1-x)-(1-x)^2 = 1-x^2$, so $f(1-x) = 1-x^2-(1-x)f(x)$. Substituting this into the original equation gives $f(x)+x(1-x^2)-x(1-x)f(x) = 2x-x^2$, that is,

$$(x^2 - x + 1)\, f(x) = x^3 - x^2 + x = x(x^2 - x + 1).$$

Since $x^2 - x + 1 = \left(x - \frac{1}{2}\right)^2 + \frac{3}{4} > 0$ for every real x, we may divide to get $f(x) = x$. Conversely $f(x) = x$ gives $x + x(1 - x) = 2x - x^2$, so it is a solution, and it is the only one.

Problem 2

Answer: $\frac{25}{12}$

The map $t \mapsto \frac{1}{1-t}$ sends $3 \mapsto -\frac{1}{2} \mapsto \frac{2}{3} \mapsto 3$, and none of these numbers is 0 or 1. Writing $A = f(3)$, $B = f\left(-\frac{1}{2}\right)$ and $C = f\left(\frac{2}{3}\right)$, the equation at $x = 3$, $x = -\frac{1}{2}$ and $x = \frac{2}{3}$ gives

$$A + B = 3, \qquad B + C = -\tfrac{1}{2}, \qquad C + A = \tfrac{2}{3}.$$

Adding the first and third and subtracting the second gives $2A = 3 + \frac{1}{2} + \frac{2}{3} = \frac{25}{6}$, so $f(3) = \frac{25}{12}$.

Such a function does exist, so the question makes sense: with $u = \frac{1}{1-x}$ and $w = \frac{x-1}{x}$, the function $f(x) = \frac{1}{2}(x - u + w)$ satisfies $f(x) + f(u) = \frac{1}{2}(x - u + w) + \frac{1}{2}(u - w + x) = x$, because the map sends u to w and w to x.

Problem 3

Answer: $f(x) = \frac{1}{3}x^3 + cx$ for a rational constant c

Note that since $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$, the function $\frac{1}{3}x^3$ satisfies the equation. Let $g(x) = f(x) - \frac{1}{3}x^3$, which also maps $\mathbb{Q}$ to $\mathbb{Q}$. Subtracting the identity for $\frac{1}{3}x^3$ from the given equation shows that $g(x + y) = g(x) + g(y)$ for all rationals x, y. By Cauchy's theorem over $\mathbb{Q}$, $g(x) = cx$ where $c = g(1)$ is rational. Hence $f(x) = \frac{1}{3}x^3 + cx$.

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Conversely, for any rational c this f maps rationals to rationals, and $f(x+y)-f(x)-f(y)=\frac{1}{3}\left[(x+y)^3-x^3-y^3\right]=xy(x+y)$, so every such function is a solution.

Problem 4

The solutions are $f(x)=x$ and $f(x)=-x$. Write $P(x,y)$ for the given equation.

$P(0,y)$ gives $f\big(f(0)\big)=f(0)f(y)$ for all y. If $f(0)\neq 0$, then $f(y)=f(f(0))/f(0)$ for all y, so f is a constant k; but then $P(x,y)$ reads $k=k^2+x$ for all real x, which is impossible. Therefore $f(0)=0$.

$P(x,0)$ now gives $f\big(f(x)\big)=x$ for all real x. In particular every real number t is a value of f, since $t=f\big(f(t)\big)$.

Let $a=f(1)$. Then $P(1,y)$ gives $f\big(f(y)+a\big)=af(y)+1$ for all y. Given any real t, choose $y=f(t)$, so that $f(y)=t$; this yields $f(t+a)=at+1$ for every real t. Writing $s=t+a$, we get $f(s)=as-a^2+1$ for all real s. Since $f(0)=0$, we need $a^2=1$, so either $a=1$ and $f(s)=s$, or $a=-1$ and $f(s)=-s$.

Finally we check both. If $f(x)=x$, both sides equal $xy+x$. If $f(x)=-x$, the left side is $f(-xy-x)=xy+x$ and the right side is $(-x)(-y)+x=xy+x$. So both functions are solutions, and there are no others.

Problem 5

The solutions are the functions $f(n)=n+a$ for even n and $f(n)=n+2-a$ for odd n, one for each odd integer a.

Applying f to both sides of $f(f(n))=n+2$ gives $f\big(f(f(n))\big)=f(n+2)$. On the other hand, the equation applied at the integer $f(n)$ gives $f\big(f(f(n))\big)=f(n)+2$. Hence $f(n+2)=f(n)+2$ for all n, and by induction in both directions $f(n+2k)=f(n)+2k$ for all integers n,k.

Let $a=f(0)$ and $b=f(1)$. Then $f(n)=n+a$ for even n and $f(n)=n-1+b$ for odd n. If a were even, we would have $f(a)=2a$, but $f(a)=f(f(0))=2$, giving $a=1$, which is not even. So a is odd, and then $f(a)=a-1+b$; since $f(a)=f(f(0))=2$, we get $b=3-a$. Therefore $f(n)=n+a$ for even n and $f(n)=n+2-a$ for odd n, with a odd.

Conversely, let a be any odd integer and define f this way. If n is even, $f(n)=n+a$ is odd, so $f(f(n))=(n+a)+2-a=n+2$. If n is odd, $f(n)=n+2-a$ is even, so $f(f(n))=(n+2-a)+a=n+2$. So every such function works. For example, $a=1$ gives $f(n)=n+1$, while $a=3$ gives $f(n)=n+3$ for even n and $f(n)=n-1$ for odd n.

Problem 6

The solutions are $f(x)=0$ and $f(x)=x$.

Putting $x=0$ gives $f(y)=f(0)^2+f(y)$, so $f(0)=0$. Putting $y=0$ gives $f(x^2)=f(x)^2$ for all x; in particular $f(t)\geq 0$ for all $t\geq 0$. Putting $y=-x^2$ gives $0=f(x)^2+f(-x^2)$, so $f(-x^2)=-f(x^2)$; that is, $f(-t)=-f(t)$ for every $t\geq 0$, and hence f is odd.

Next we show that f is additive. For $t\geq 0$ and any y, putting $x=\sqrt{t}$ gives $f(t+y)=f(\sqrt{t})^2+f(y)=f(t)+f(y)$. For $t<0$, let $s=-t>0$; applying what we have just proved to s and $t+y$ gives $f(y)=f\big((t+y)+s\big)=f(t+y)+f(s)$, so $f(t+y)=f(y)-f(s)=f(y)+f(t)$ by oddness. So $f(t+y)=f(t)+f(y)$ for all real t,y.

An additive function that is non-negative on $[0,\infty)$ is of the form $f(x)=cx$: for $a<b$ we have $f(b)=f(a)+f(b-a)\geq f(a)$, so f is increasing; additivity gives $f(q)=cq$ for rational q, where $c=f(1)$, and for real x and rationals $p<x<q$ we get $cp\leq f(x)\leq cq$, so letting $p,q\to x$ gives $f(x)=cx$.

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Finally $f(x^2) = f(x)^2$ becomes $cx^2 = c^2x^2$, so $c = 0$ or $c = 1$. Both work: $f(x) = 0$ gives $0 = 0$, and $f(x) = x$ gives $x^2 + y = x^2 + y$.

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